

ALEMU SISAY NIGRU

AI Research Scientist • PhD, AI in Medicine • Deep Learning for Medical Imaging

📍 Brescia, Italy

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🌐 alexisisay.github.io/alemu-portfolio

PROFESSIONAL SUMMARY

PhD in AI in Medicine at the University of Brescia with 5+ years of hands-on experience in deep learning for medical imaging and AI-driven clinical decision support. Specialized in spine MRI analysis, transformer-based architectures (Swin3D), and fine-tuning LLMs for automated radiological report generation. Proven ability to translate cutting-edge AI research into deployed clinical tools — including LumbarAI, a live MRI analysis platform currently used at Poliambulatorio Oberdan. International research experience spanning Italy, the USA (NYU), and Ethiopia (CDHI). Seeking postdoctoral or junior AI scientist roles in medical imaging, healthcare AI, or clinical NLP.

PROFESSIONAL EXPERIENCE

AI Consultant | Xray Service Srl, Brescia, Italy

Dec 2025 – Present

- Developed and deployed **LumbarAI**, a Swin3D Transformer-based MRI analysis platform automating grading of 11 radiological features across the lumbar spine — achieving a 75–85% reduction in active radiologist reporting time.
- Integrated PACS/DICOM workflows to automate image retrieval and sequence validation, reducing physician data-entry latency.
- Engineered a Human-in-the-Loop (HitL) framework with real-time XAI explainability heatmaps, ensuring 100% clinician oversight of all AI predictions.
- Designed a SQLite-backed audit and correction system to log radiologist edits and build a 'Gold Standard' feedback loop for continuous model fine-tuning.
- Built a bilingual (EN/IT) LLM-powered report generation engine enabling immediate structured clinical documentation and patient risk screening.

Visiting PhD Researcher | New York University (NYU) – New York, USA

Apr 2025 – Oct 2025

- Fine-tuned Large Language Models (LLMs) for automated radiological report generation, improving reporting efficiency and linguistic quality on clinical spine MRI datasets.
- Developed and benchmarked machine learning algorithms for low back pain (LBP) severity grading, achieving robust generalization across heterogeneous MRI scanners.
- Conducted external validation of SpineNetV2 across diverse patient demographics and MRI equipment, contributing to its clinical translation.
- Collaborated with multidisciplinary teams of radiologists, spine surgeons, and computer vision engineers on model interpretability and clinical validation.

Visiting PhD Researcher | Center for Digital Health & Implementation Science (CDHI), Ethiopia Sep 2024 – Dec 2024

- Contributed to the design and deployment of AI diagnostic tools integrated into Electronic Medical Record (EMR) systems in resource-constrained clinical environments.
- Developed graphical user interfaces (Streamlit/Gradio) to facilitate seamless adoption of deep learning models by clinical staff.
- Adapted AI model APIs for low-resource hardware and variable connectivity conditions, supporting real-world AI deployment in global health settings.

Extraordinary Internship | Poliambulatorio Oberdan, Brescia, Italy

May 2022 – Dec 2022

- Acquired and analyzed digital medical images for diagnostic activities; applied deep learning methods for MRI image segmentation and analysis.

Assistant Lecturer | University of Gondar – Dept. of Electrical & Computer Engineering, Gondar, Ethiopia

Aug 2016 – Aug 2019

- Taught undergraduate courses: C++ Programming, Digital Logic Design, Circuit Theory, Signal Analysis, and Communication Systems.
- Served as Chairperson of the Communication Engineering focus area, overseeing 100+ students and 30+ lecturers.
- Led community-based technology transfer and capacity-building projects.

EDUCATION

Ph.D. in Artificial Intelligence in Medicine | [University of Brescia, Italy](#) December 2022 – May 2026
Dissertation: AI-driven clinical decision support for personalized management of low back pain.

M.Sc. in Communication Technologies and Multimedia | [University of Brescia, Italy](#) 2019 – 2022
Final Grade: 110/110 cum Laude • Thesis: Deep learning–based neuroimage retrieval
Erasmus+ Mobility: Mid Sweden University (2020–2021) — Sensor Networks, Embedded Systems, Business English.

B.Sc. in Electrical and Computer Engineering | [University of Gondar, Ethiopia](#) 2012 – 2016
Final Grade: 3.89/4.00 — Gold Medalist, 1st in Graduating Class

TECHNICAL SKILLS

AI & Deep Learning	PyTorch, TensorFlow, Swin3D Transformers, CNNs, Transfer Learning, Domain Adaptation, Self-supervised Learning
LLMs & NLP	Fine-tuning LLMs, Automated Report Generation, Clinical NLP, Prompt Engineering
Medical Computing	DICOM, PACS, NiftI, Nibabel, Nipype, MRI/Neuroimage Processing, PyDICOM
Explainable AI	Grad-CAM, XAI heatmaps, Human-in-the-Loop (HitL) Frameworks
Software & MLOps	Python (Advanced), C++, SQL, Git, Docker, ETL Pipelines, Streamlit, Gradio
Clinical Research	Statistical Methods, MRI Severity Grading, PROMs, External Validation, IRB-compliant workflows
Languages	English (C1), Italian (B1), Amharic (native)

KEY PROJECTS

LumbarAI — MRI Clinical Analysis Platform
End-to-end deployed Swin3D Transformer system for automated grading of 11 lumbar spine radiological features. Integrated with PACS/DICOM, XAI heatmaps, HitL correction framework, bilingual report generation, and SQLite audit trail. Currently operational at Poliambulatorio Oberdan, Brescia.

SpineNetV2 — External Validation & LBP Severity Grading (NYU)
Validated SpineNetV2 on a comprehensive multi-site dataset and developed ML algorithms for LBP severity grading. Resulted in a peer-reviewed publication in NASSJ (2024) and ongoing multimodal patient stratification work.

Multimodal Risk Stratification in Spine Health
Integrated radiological severity scores with psychosocial PROMs (WHOQOL, Roland-Morris, Pain NRS, COPE-NVI) and unsupervised clustering to identify patient risk profiles and support personalized treatment strategies.

Infant Brain MRI Segmentation (Leveraging Adult Priors)
Co-developed a novel approach for enhancing infant brain MRI segmentation by transferring learned representations from adult brain priors, addressing data scarcity in neonatal neuroimaging.

PUBLICATIONS

Peer-Reviewed

- ▶ Nigru, A. S., Benini, S., Bonetti, M., Bragaglio, G., Frigerio, M., Maffezzoni, F., Leonardi, R. "External validation of SpineNetV2 on a comprehensive set of radiological features for grading lumbosacral disc pathologies." North American Spine Society Journal (NASSJ), 2024. doi: 10.1016/j.xnsj.2024.100564

Under Review / Preprints

- ▶ Nigru, A. S., et al. "LumbarAI: A Swin3D Transformer-Based Framework for Computer-Aided Diagnosis of Lumbar Spine Disorders from Sagittal MRI." (Under review, 2026)
- ▶ Nigru, A. S., Benini, S., Wang, Y., et al. "Toward a Clinically Integrated AI Framework for Personalized Spine Care." SSRN Preprint, 2026. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5860785
- ▶ Nigru, A. S., et al. "Unsupervised Clustering of Psychosocial Profiles for Patient Stratification in Spinal Pain Management." (Under review, 2026)
- ▶ Nigru, A. S., Svanera, M., Dibble, A., Dalby, C., et al. "Rewiring Development: Leveraging Adult Brain Priors for Enhancing Infant Brain MRI Segmentation." arXiv, 2025. <https://arxiv.org/abs/2510.09306>
- ▶ Nigru, A. S., et al. "Integrating Radiological Severity and Patient-Reported Outcomes for Risk Factor Identification in Spine Health." (Under review, 2026)

CERTIFICATES & TRAINING

🎓 Oxford ML Summer School — AI for Global Goals

🎓 IEEE ComSoc eHealth TC PhD School — IEEE Communication Society

🎓 Higher Diploma in Teaching Methods — University of Gondar

🎓 Applied Data Science Lab — WorldQuant University (6 months)

🎓 Statistical Methods & Data Analysis in Medical Research — University of Brescia PhD School

🎓 Multiple short courses in ML, Deep Learning, and AI

HONOURS & AWARDS

🏆 Gold Medal — 1st in BSc Graduating Class, University of Gondar (2016)

🏆 Appreciation Letter — Ethio telecom, for Automating SIM Card Profile Management System

🏆 3rd Place — Ethiopian Space Science & Technology Q&A Competition (7 universities)

🎓 MSc — 110/110 cum Laude, University of Brescia (2022)